

1 mark

Input

5
1 3 5 7 9
4
2 4 6 8

Output

1 2 3 4 5 6 7 8 9

```

1 #include <stdio.h>
2 int main() {
3     int n, m;
4     int i, j, k;
5     printf("Enter size of first array: ");
6     scanf("%d", &n);
7     int a[n];
8     printf("Enter elements of first array:\n");
9     for (i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12     printf("Enter size of second array: ");
13     scanf("%d", &m);
14     int b[m];
15     printf("Enter elements of second array:\n");
16     for (j = 0; j < m; j++) {
17         scanf("%d", &b[j]);
18     }
19     int c[n + m];
20     i = 0; j = 0; k = 0;
21     while (i < n && j < m) {
22         if (a[i] <= b[j]) {
23             c[k++] = a[i++];
24         } else {
25             c[k++] = b[j++];
26         }
27     }
28     while (i < n) {
29         c[k++] = a[i++];
30     }
31     while (j < m) {
32         c[k++] = b[j++];
33     }
}

```

Submitted

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QUESTIONS

Q1
Searching
1 mark

Q2
Array
1 mark

Q3
Merging
1 mark

3/3 answered

Q2 Searching1 mark

Write a C program for Count the Occurrences of an Element

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, key, count = 0;
5     printf("Enter number of elements: ");
6     scanf("%d", &n);
7
8     int arr[n];
9
10    printf("Enter array elements: ");
11    for (i = 0; i < n; i++) {
12        scanf("%d", &arr[i]);
13    }
14
15    printf("Enter element to count: ");
16    scanf("%d", &key);
17
18    for (i = 0; i < n; i++) {
19        if (arr[i] == key)
20            count++;
21    }
22
23    printf("Occurrences of %d = %d\n", key, count);
24
25    return 0;
26 }
```

Submitted

Input

```
5
1 2 3 2 2
2
```

Output

```
Occurrences of 2 = 3
```

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QUESTIONS

Q1

Searching

1 mark

✓

Q2

Array

1 mark

✓

Q3

Merging

1 mark

✓

3/3 answered

Q1 Array

1 mark

Write a program for binary search

Your Code

```
1 #include <stdio.h>
2 int binarySearch(int arr[], int n, int target) {
3     int low = 0, high = n - 1;
4     while (low <= high) {
5         int mid = low + (high - low) / 2;
6         if (arr[mid] == target)
7             return mid;
8         else if (arr[mid] < target)
9             low = mid + 1;
10        else
11            high = mid - 1;
12    }
13    return -1;
14 }
15 int main() {
16     int n, target;
17     printf("Enter number of elements: ");
18     scanf("%d", &n);
19     int arr[n];
20     printf("Enter sorted array elements: ");
21     for (int i = 0; i < n; i++)
22         scanf("%d", &arr[i]);
23     printf("Enter target element: ");
24     scanf("%d", &target);
25     int result = binarySearch(arr, n, target);
26     if (result != -1)
27         printf("Element found at index %d\n", result);
28     else
29         printf("Element not found\n");
30     return 0;
31 }
```

Submitted

Input

5
10 20 30 40 50

Output

Element found at index 2